浙江大學
ZHEJIANG UNIVERSITY



信号与系统
实验报告

实验名称	lab4
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1 Problem 1

1.1 (a)

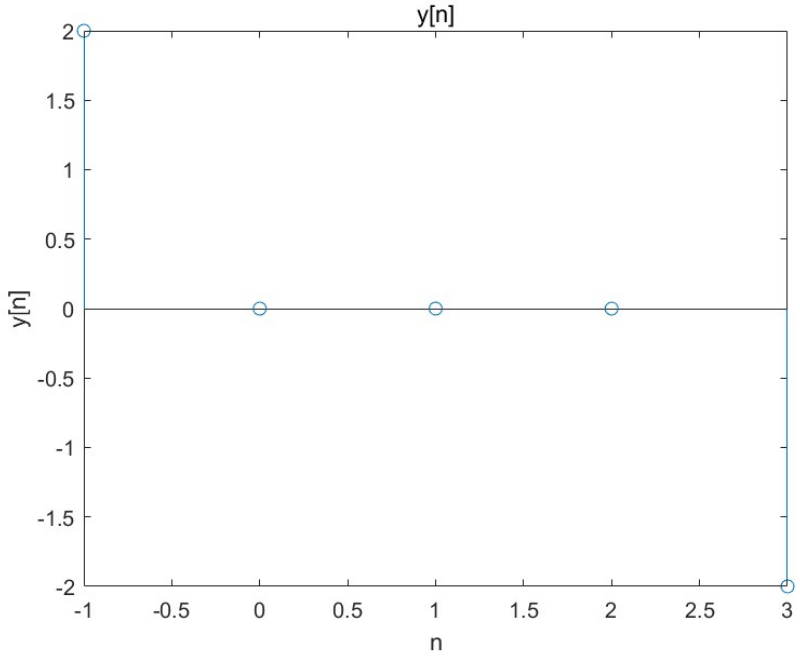


Figure 1: Answer to problem 1 (a)

1.2 (b)

$$y[n] = h[n]*x[n] = \sum_{k=-\infty}^{+\infty} h[k]x[n-k] = \delta[n-a-c] + \delta[n-a-d] + \delta[n-b-c] + \delta[n-b-d]$$

So $y[n]$ is nonzero on the interval$[a+c, b+d]$. When $a = 0, b = N-1, c = 0, d = M-1$, $n_y = [0 : M+N-2]$which is coincident with the expected result.

1.3 (c)

Theoretical calculated value is: $y[n] = h[n] * x[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k] = 2 - \left(\frac{1}{2}\right)^n$

Since $x[n]$ is pegged in range of $0 \leq n \leq 24$ and $h[n]$ is pegged in range of $0 \leq n \leq 14$,the output for truncated signal:

$$y'[n] = \sum_{k=\max\{2,n-14\}}^{\min\{24,n\}} \left(\frac{1}{2}\right)^{k-2}$$

The images of the two signals are as follows.

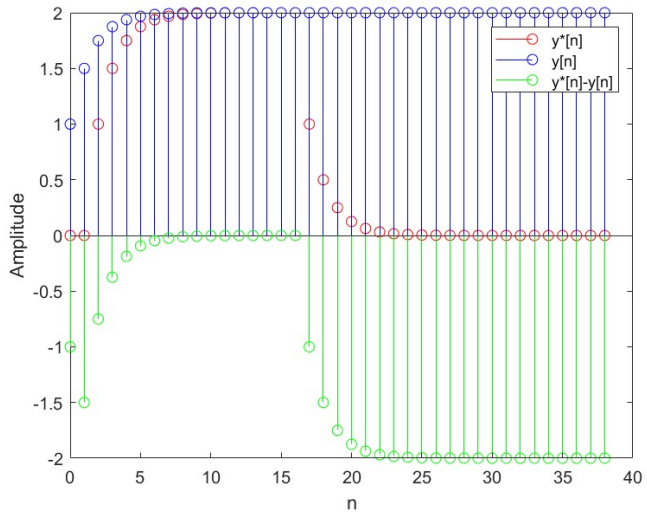


Figure 2: Answer 1 (c)

We can see from the image that $y'[n]$ is obtained by shifting $y[n]$ to the right by two time units within the range of $2 \leq n \leq 16$. The red curve represents the difference between $y[n]$ and $y'[n]$.The value of $y'[n]$ is nearly correct when $8 \leq n \leq 16$

1.4 (d)

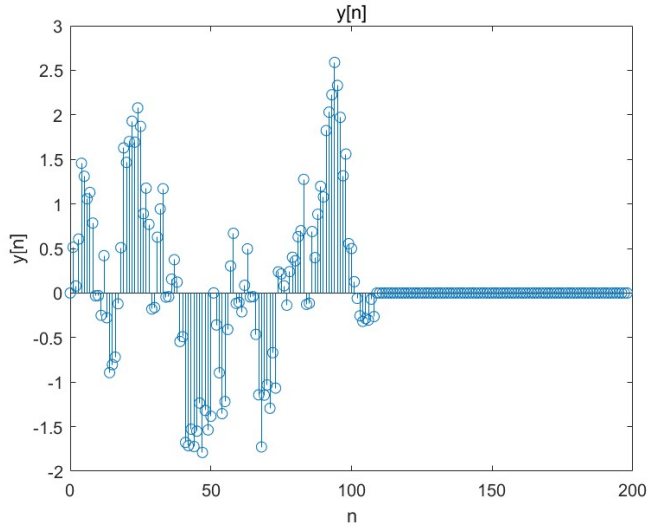


Figure 3: Answer to problem 1 (d)

1.5 (e)

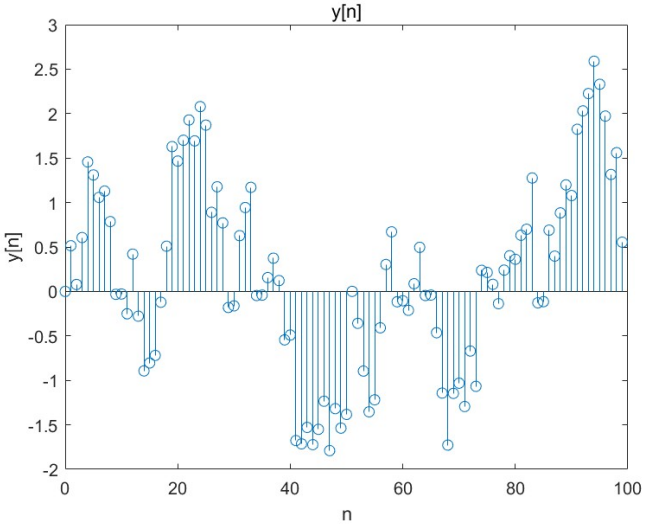


Figure 4: Answer to problem 1 (e)

We can see it is the same as the result in (d)

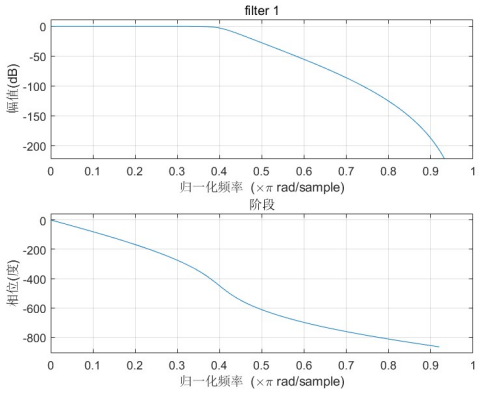
2 Problem 2

2.1 (a)

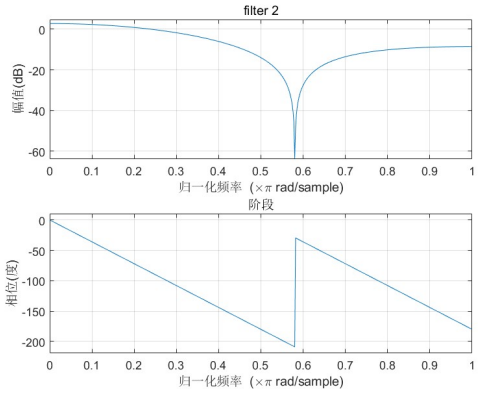
```
>> wc=0.4;
n1=10;n2=4;n3=12;
[b1,a1]=butter(n1,wc);
a2=1;b2=remez(n2,[0 wc-0.04 wc+0.04 1],[1 1 0 0]);
a3=1;b3=remez(n3,[0 wc-0.04 wc+0.04 1],[1 1 0 0]);
```

Figure 5: Answer to Problem 2 (a)

2.2 (b)



(a) Answer to problem 2(b)1



(b) Answer to problem 2(b)2

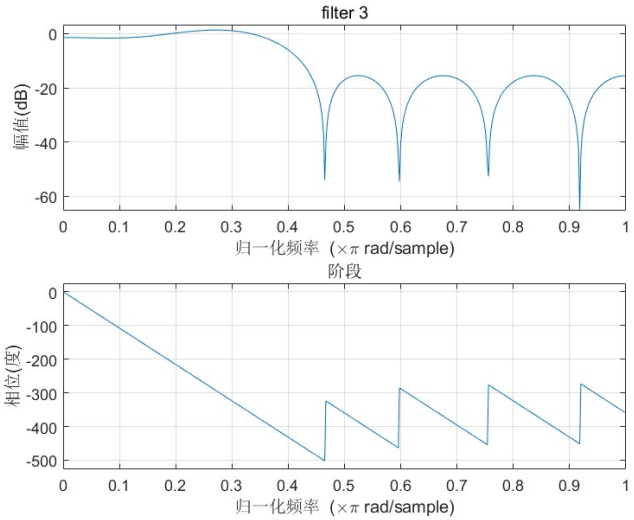
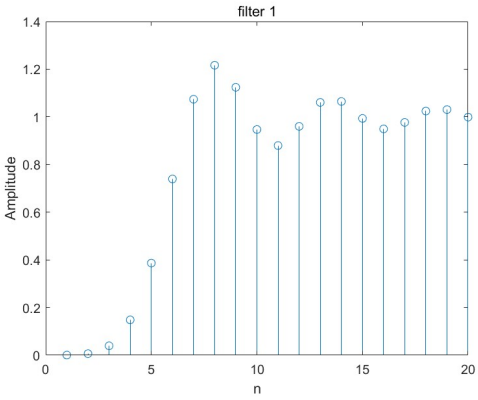


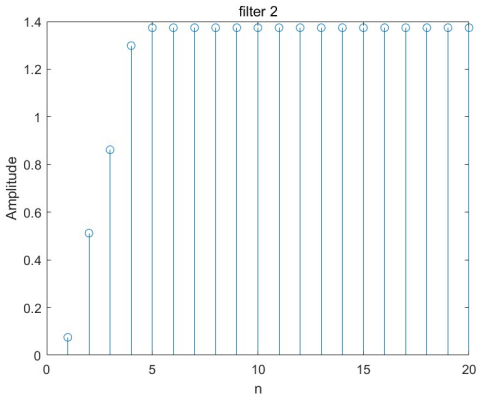
Figure 7: Answer to problem 2(b)3

We can see that the ω_c is basically the cutoff frequency of the filter. Filter 2 and filter 3 have linear phase.

2.3 (c)



(a) Answer to problem 2(c)1



(b) Answer to problem 2(c)2

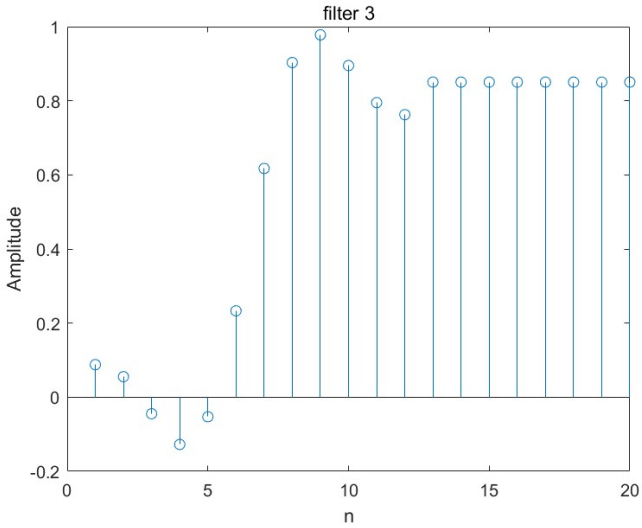


Figure 9: Answer to problem 2(c)3

It can be seen that Filter 1 has a significant overshoot.

2.4 (d)

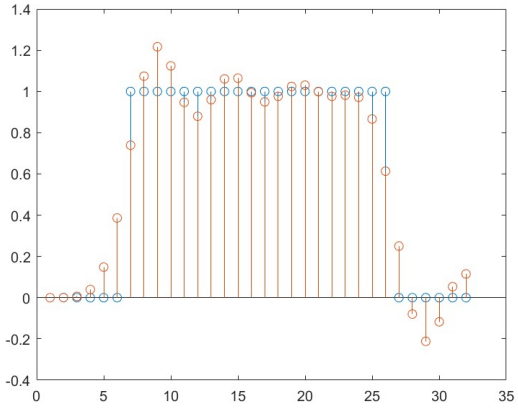


Figure 10: Answer to problem 2(d)

It is easy to see that the discontinuities in x16 line up with the smoothed discontinuities in y16。

2.5 (e)

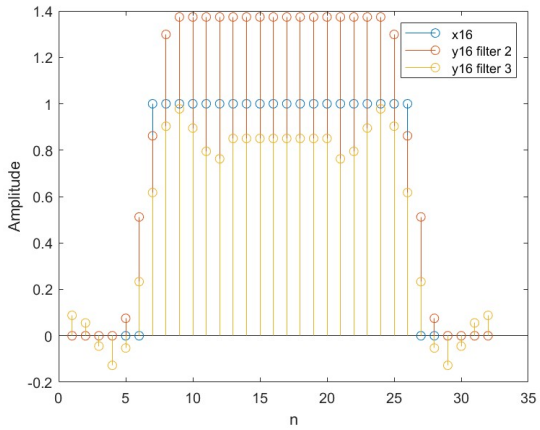


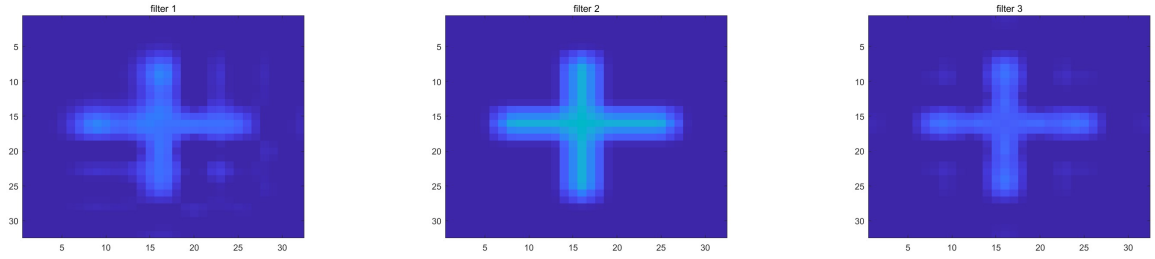
Figure 11: Answer to problem 2(d)

Similarly, we can image the picture of filter two and three.

2.6 (f)

The answer to this question is provided in the code section below

2.7 (g)



(a) Answer to problem 2(g)1 (b) Answer to problem 2(g)2 (c) Answer to problem 2(g)3

Figure 12: Answers to problem 2(g)

2.8 (h)

Filter 1 exhibits more distortion, attributable to its non-linear phase.

3 Problem 3

3.1 (a)

A signal x[n] is purely real if its DTFS coefficients satisfy conjugate symmetry, i.e., $a_k = a_{-k}^*$

3.2 (b)

We can obtain that:

$$a_0 = 1, a_1 = 2e^{-j\frac{\pi}{3}}, a_2 = e^{j\frac{\pi}{4}}, a_3 = e^{-j\frac{\pi}{4}}, a_4 = 2e^{j\frac{\pi}{3}}$$

So the vector is :

$$a = [1, 2e^{-j\frac{\pi}{3}}, e^{j\frac{\pi}{4}}, e^{-j\frac{\pi}{4}}, 2e^{j\frac{\pi}{3}}]$$

3.3 (c)

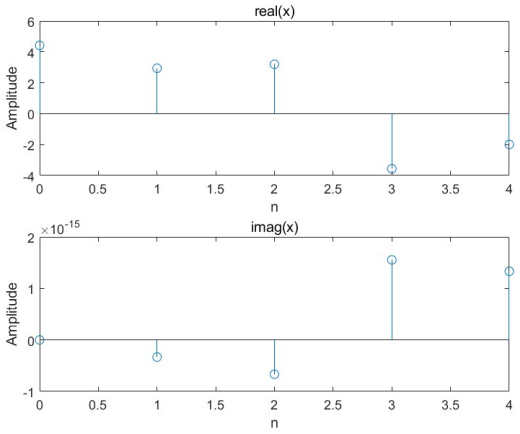


Figure 13: Answer to problem 3(c)

Because of the roundoff errors there exits a very small nonzero imaginary part in x[n], which is negligible.

3.4 (d)

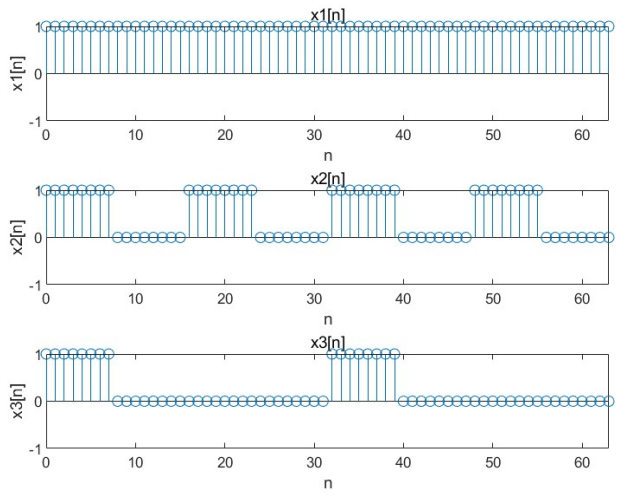


Figure 14: Answer to problem 3(d)

3.5 (e)

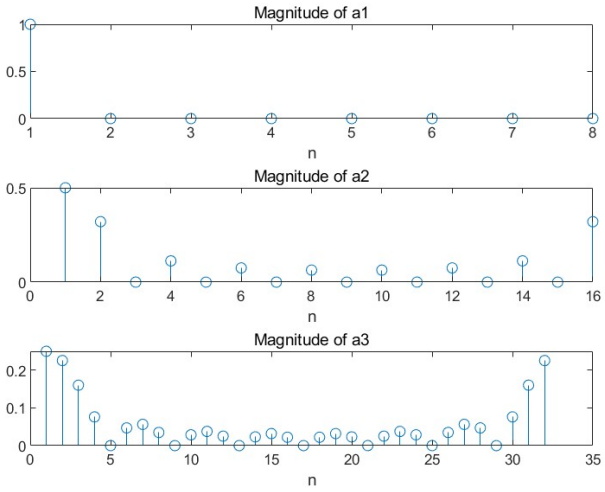


Figure 15: Answer to problem 3(e)

The DC component of a signal is the average of the signal over a period,thus

$$a_1(1) = 1 \quad a_2(1) = \frac{1}{2} \quad a_3(1) = \frac{1}{4}$$

We can see this on the diagram as well

3.6 (f)

Since $a_k = a_{k+32}$,we can determine $a3(18).....a3(32)$correspond to $a_{15}.....a_1$

3.7 (g)

According to the condition for $x_{3_all}[n]$:

$$x_{3_all}[n] = \sum_{k=-15}^{16} a_k e^{jk(2\pi/32)n}$$

The conjugation of $x_{3_all}[n]$ is:

$$x^*_{3_all}[n] = \left(\sum_{k=-15}^{16} a_k e^{jk(2\pi/32)n} \right)^* = \sum_{k=-15}^{16} a^*_k e^{-jk(2\pi/32)n}$$

Since the original signal $x_3[n]$ (from which a_k are derived) is purely real, its Fourier Series (FS) coefficients a_k are conjugate symmetric. That is, $a_k = a^*_{-k}$, or equivalently, $a^*_k = a_{-k}$.

Substituting k with $-k'$ (let $k = -k'$, so when k goes from -15 to 16, $-k'$ also covers a range that can be re-indexed) in the summation for $x^*_{3_all}[n]$: When $k = -15$, $k' = 15$. When $k = 16$, $k' = -16$. So the sum becomes over k' from 15 down to -16.

$$x^*_{3_all}[n] = \sum_{k'=-16}^{15} a^*_{-k'} e^{-j(-k')(2\pi/32)n} = \sum_{k'=-16}^{15} a^*_{-k'} e^{jk'(2\pi/32)n}$$

Using the conjugate symmetry $a^*_{-k'} = (a_{-(-k')}) = a_{k'}$:

$$x^*_{3_all}[n] = \sum_{k'=-16}^{15} a_{k'} e^{jk'(2\pi/32)n}$$

Renaming the summation index k' back to k:

$$x^*_{3_all}[n] = \sum_{k=-16}^{15} a_k e^{jk(2\pi/32)n}$$

The problem statement in the image presents a slightly different final step involving the property $a_{16} = a_{-16}$ for a real signal with period $N = 32$ (since $a_{N/2}$ must be real if N is even, and $a_{N/2} = a^*_{-N/2}$, it implies $a_{N/2} = a_{-N/2}$ if $a_{N/2}$ is real. Also, generally for real signals, $a_k = a_{k+N}$). The summation range for $x_{3_all}[n]$ is from $k = -15$ to $k = 16$. This range includes $N = 32$ unique terms. The summation for $x^*_{3_all}[n]$ derived above is $\sum_{k=-16}^{15} a_k e^{jk(2\pi/32)n}$. These two sums are over the same set of $N = 32$ unique frequency components due to the periodicity of a_k with period $N = 32$. Specifically, $a_{-16} = a_{-16+32} = a_{16}$. So, the term for $k = -16$ in the sum for $x^*_{3_all}[n]$ is $a_{-16}e^{j(-16)(2\pi/32)n} = a_{16}e^{j(16)(2\pi/32)n}$ (since $e^{j(-16)\theta} = e^{j(16)\theta}$ if $\theta = \pi n$). Thus, the sum $\sum_{k=-16}^{15} a_k e^{jk(2\pi/32)n}$ is indeed equivalent to $\sum_{k=-15}^{16} a_k e^{jk(2\pi/32)n}$.

Therefore,

$$x^*_{3_all}[n] = \sum_{k=-15}^{16} a_k e^{jk(2\pi/32)n} = x_{3_all}[n]$$

Since $x^*_{3_all}[n] = x_{3_all}[n]$, the signal $x_{3_all}[n]$ must be real.

3.8 (h)

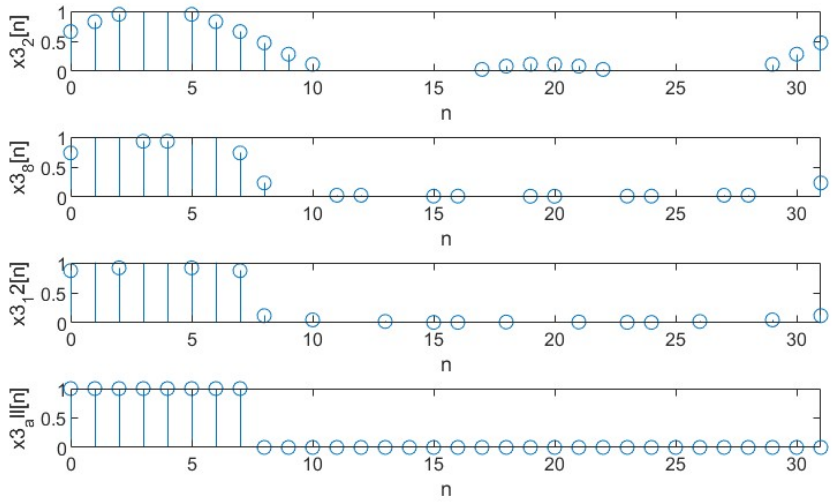


Figure 16: Enter Caption

Figure 23, which presumably shows plots of $x_{3_2}[n]$, $x_{3_8}[n]$, $x_{3_12}[n]$, and $x_{3_all}[n]$ (where the subscript indicates the extent of DTFS coefficients used, e.g., $k = -M$ to M, and $x_{3_all}[n]$ uses $k = -15$ to 16), demonstrates the convergence to $x_3[n]$ and the presence of Gibb's phenomenon.

The four figures illustrate the presence of Gibb's phenomenon in all finite-term approximations of $x_3[n]$ ($x_{3_2}[n]$, $x_{3_8}[n]$, and $x_{3_12}[n]$). These approximations, formed by summing a limited number of DTFS coefficients, exhibit characteristic overshoots and oscillations near the discontinuities of the square wave.

- The approximation using the fewest terms (referred to as the "2-term approximation", likely corresponding to $x_{3_2}[n]$ synthesized with $k \in [-2, 2]$) shows severe waveform distortion. This includes large oscillations at the discontinuities (n=0, n=7/8, n=31/0 transitions) and noticeable ripples across the regions that should be flat (i.e., where $x_3[n]$ is 0 or 1).

- The approximations using more terms (referred to as "8-term" and "12-term approximations", likely corresponding to $x_{3_8}[n]$ with $k \in [-8, 8]$ and $x_{3_12}[n]$ with $k \in [-12, 12]$ respectively) show improved flatness in the constant regions compared to the 2-term case. However, they still display persistent oscillations (overshoots and undershoots) near the discontinuities, which are characteristic of the Gibb's

phenomenon. As more terms are included, these oscillations become more compressed towards the discontinuities but their peak amplitude does not diminish to zero.

Thus, the synthesis of this discrete-time square wave with a finite number of DTFS coefficients clearly displays the Gibb's phenomenon.

Conversely, the full-term reconstruction, $x_{3_all}[n]$ (synthesized using all $N_3 = 32$ unique DTFS coefficients, corresponding to $k = -15$ to 16), perfectly reproduces the original square wave $x_3[n]$ without any observable Gibb's artifacts. This complete disappearance of the Gibb's phenomenon in the full-term reconstruction (which is, by definition, the original signal $x_3[n]$ within numerical precision) validates the completeness of DTFS theory for representing periodic discrete-time signals. The signal $x_{3_all}[n]$ is equal to the original vector representing one period of $x_3[n]$ within the bounds of MATLAB's roundoff error.

4 Code

4.1 Problem 1

Listing 1: matlab code

```
%Problem(a)
clear;
clc;

x=[1 0 1]; % input and impulse response
h=[2 0 -2];
y=conv(h,x); % convolution
ny=-1:3;
stem(ny,y);
title('y[n]');
xlabel('n');
ylabel('y[n]');

%Problem(c)
clear;
clc;

nx=0:24;
```

```
19  x=(0.5.^(nx-2)).*[zeros(1,2) ones(1,23)]; % x[n]
20
21  nh=0:14;
22  h=ones(1,15); % h[n]
23
24  y=conv(h,x);
25  ny=0:38;
26  stem(ny,y,'r'); % truncated output
27
28  hold on;
29  stem(ny,2-2.^(-ny),'b'); % analytical output
30
31  hold on;
32  stem(ny,y-2+2.^(-ny),'g'); % difference between the two outputs
33  xlabel('n');
34  ylabel('Amplitude');
35  legend('y*[n]','y[n]','y*[n]-y[n]');
36
37  %Problem(d)
38  clear;
39  clc;
40
41  nx=0:99;
42  x=cos(nx.^2).*sin(2*pi/5.*nx); % x[n]
43
44  nh=0:99;
45  h=0.9.^nh.*(nh>=0 & nh<=9); % h[n]
46
47  y=conv(h,x);
48  ny=0:198; % range for y[n]
49  stem(ny,y);
50  xlabel('n');
51  ylabel('y[n]');
52  title('y[n]');
53
54  %Problem (e)
55  clear;
56  clc;
```

```
57
58  n=0:99;
59  h=0.9.^n.*(n>=0&n<10); % h[n]
60
61  nx0=0:49;
62  x0=cos(nx0.^2).*sin(2*pi/5.*nx0); % x0[n]
63  nx1=nx0+50;
64  x1=cos(nx1.^2).*sin(2*pi/5.*nx1); % x1[n]
65
66  y0=conv(h,x0); % y0[n]
67  y1=conv(h,x1); % y1[n]
68  y=[y0 zeros(1,50)]+[zeros(1,50) y1]; % y0[n]+y1[n-50]
69  stem(n,y(1:100));
70  xlabel('n');
71  ylabel('y[n]');
72  title('y[n]');
```

4.2 Problem 2

Listing 2: matlab code

```
1   %Problem (b)
2   clear;
3   clc;
4
5   wc=0.4;
6   n1=10;n2=4;n3=12;
7   [b1,a1]=butter(n1,wc);
8   a2=1;b2=remez(n2,[0 wc-0.04 wc+0.04 1],[1 1 0 0]);
9   a3=1;b3=remez(n3,[0 wc-0.04 wc+0.04 1],[1 1 0 0]);
10
11  figure; % filter_1
12  freqz(b1,a1);
13  title('filter 1');
14
15  figure; % filter_2
16  freqz(b2,a2);
17  title('filter 2');
```

```
18
19  figure;% filter_3
20  freqz(b3,a3);
21  title('filter 3');
22
23  %Problem (c)
24  clear;
25  clc;
26
27  wc=0.4;
28  n1=10;n2=4;n3=12;
29  [b1,a1]=butter(n1,wc);
30  a2=1;b2=remez(n2,[0 wc-0.04 wc+0.04 1],[1 1 0 0]);
31  a3=1;b3=remez(n3,[0 wc-0.04 wc+0.04 1],[1 1 0 0]);
32
33  n=1:20; % the input signal
34  x=ones(1,20);
35  h1=filter(b1,a1,x); % the step responses
36  h2=filter(b2,a2,x);
37  h3=filter(b3,a3,x);
38
39  figure;
40  stem(n,h1);
41  xlabel('n');
42  ylabel('Amplitude');
43  title('filter 1');
44
45  figure;
46  stem(n,h2);
47  xlabel('n');
48  ylabel('Amplitude');
49  title('filter 2');
50
51  figure;
52  stem(n,h3);
53  xlabel('n');
54  ylabel('Amplitude');
55  title('filter 3');
```

```
56
57  %Problem (d)
58  load plus;
59
60  wc = 0.4;
61  n1 = 10; n2 = 4; n3 = 12;
62  [b1,a1] = butter(n1,wc);
63  a2 = 1; b2 = remez(n2,[0 wc-0.04 wc+0.04 1],[1 1 0 0]);
64  a3 = 1; b3 = remez(n3,[0 wc-0.04 wc+0.04 1],[1 1 0 0]);
65
66  x16 = x(:,16); % the column input
67  y16 = filter(b1,a1,[x16;zeros(n1/2,1)]); % filter response
68
69
70  stem(x16);
71  hold on
72  stem(y16(n1/2+1:end));
73
74  %Problem (e)
75  clear;
76  clc;
77
78  load plus;
79
80  wc=0.4;
81  n1=10;n2=4;n3=12;
82  [b1,a1]=butter(n1,wc);
83  a2=1;b2=remez(n2,[0 wc-0.04 wc+0.04 1],[1 1 0 0]);
84  a3=1;b3=remez(n3,[0 wc-0.04 wc+0.04 1],[1 1 0 0]);
85
86  x16=x(:,16);
87  y16_2=filter(b2,a2,[x16;zeros(n2/2,1)]); % filter 2's response
88  y16_3=filter(b3,a3,[x16;zeros(n3/2,1)]); % filter 3's response
89
90  figure;
91  stem(x16);
92  hold on;
93  stem(y16_2(n2/2+1:end));
```

```
94  hold on;
95  stem(y16_3(n3/2+1:end));
96  xlabel('n');
97  ylabel('Amplitude');
98  legend('x16','y16 filter 2','y16 filter 3');
99
100 %Problem (f)
101 function y = filt2d(b,a,d,x)
102  %d = n/2;n is the order of the 1D filter
103  %a b is the coefficients of the 1Dfilter
104  %size fo x y z is N*N
105
106  [m,n] = size(x);
107  y = zeros(m, n);
108  z = zeros(m, n);
109
110  for i = 1:n  %each column
111      xi = x(:,i);
112      z1 = filter(b,a,[xi;zeros(d,1)]);
113      z(:,i) = z1(d+1:end);
114  end
115
116  for j = 1:m  %each row
117      zj = z(j,:);
118      y1 = filter(b,a,[zj zeros(1,d)]);
119      y(j,:) = y1(d+1:end);
120  end
121
122 end
```

4.3 Problem 3

Listing 3: matlab code

```
1  %Problem (c)
2  clear;
3  clc;
4
```

```
a = [1 2*exp(-1j*pi/3) exp(1j*pi/4) exp(-1j*pi/4) 2*exp(1j*pi/3)];
x = zeros(1,5);
nx = 0:4;
for n = 1:5
   for k = 1:5
      x(n) = x(n)+a(k)*exp(1j*(k-1)*2*pi/5*(n-1)); % compute x[n]
   end
end
xr = real(x); % real and imaginary part
xi = imag(x);

subplot(2,1,1)
stem(nx,xr);
xlabel('n');
ylabel('Amplitude');
title('real(x)');
subplot(2,1,2)
stem(nx,xi);
xlabel('n');
ylabel('Amplitude');
title('imag(x)');

%Problem (d)
clear;
clc;

x1 = ones(1,8); % each periodic signal
x2 = [ones(1,8),zeros(1,8)];
x3 = [ones(1,8),zeros(1,24)];
n = 0:63;

x1_0 = [x1 x1 x1 x1 x1 x1 x1 x1];
x2_0 = [x2 x2 x2 x2];
x3_0 = [x3 x3]; % repeat the vectors

subplot(3,1,1);
stem(n,x1_0);
xlabel('n');
```

```
ylabel('x1[n]');
title('x1[n]');
axis([0 63 -1 1]);

subplot(3,1,2);
stem(n,x2_0);
xlabel('n');
ylabel('x2[n]');
title('x2[n]');
axis([0 63 -1 1]);

subplot(3,1,3);
stem(n,x3_0);
xlabel('n');
ylabel('x3[n]');
title('x3[n]');
axis([0 63 -1 1]);

%Problem (e)
clear;
clc;

x1=ones(1,8);
x2=[ones(1,8),zeros(1,8)];
x3=[ones(1,8),zeros(1,24)];

% use fft to compute coefficients
a1=1/8*fft(x1);
a2=1/16*fft(x2);
a3=1/32*fft(x3);

a1_s=abs(a1); % the magnitude
a2_s=abs(a2);
a3_s=abs(a3);

subplot(3,1,1)
stem(a1_s);
xlabel('n');
```

```
81  title('Magnitude of a1');
82
83  subplot(3,1,2)
84  stem(a2_s);
85  xlabel('n');
86  title('Magnitude of a2');
87
88  subplot(3,1,3)
89  stem(a3_s);
90  xlabel('n');
91  title('Magnitude of a3');
92
93  %Problem (h)
94  clear;
95  clc;
96
97  x3 = [ones(1,8),zeros(1,24)];
98  a3 = 1/32*fft(x3); % the FTFS coefficients
99  nx = 0:31;
100
101 x3_2 = zeros(1,32); % preset the memory
102 x3_8 = zeros(1,32);
103 x3_12 = zeros(1,32);
104 x3_all = zeros(1,32);
105
106 for n = 1:32 % x3_2
107     for k = 31:32 % the negative parts
108       x3_2(n) = x3_2(n)+a3(k)*exp(1j*(k-33)*2*pi/32*n);
109     end
110     for k = 1:3 % the positive parts
111        x3_2(n) = x3_2(n)+a3(k)*exp(1j*(k-1)*2*pi/32*n);
112     end
113 end
114 x3_2 = circshift(x3_2,1);
115 x3_2r = real(x3_2); % ignore the very small imainary part
116
117 for n = 1:32 % x3_8
118     for k = 25:32 % the negative parts
```

```
      x3_8(n) = x3_8(n)+a3(k)*exp(1j*(k-33)*2*pi/32*n);
    end
    for k=1:9 % the positive parts
        x3_8(n) = x3_8(n)+a3(k)*exp(1j*(k-1)*2*pi/32*n);
    end
end
x3_8 = circshift(x3_8,1);
x3_8r = real(x3_8);

for n = 1:32 % x3_12
    for k = 21:32 % the negative parts
      x3_12(n) = x3_12(n)+a3(k)*exp(1j*(k-33)*2*pi/32*n);
    end
    for k = 1:13 % the positive parts
        x3_12(n) = x3_12(n)+a3(k)*exp(1j*(k-1)*2*pi/32*n);
    end
end
x3_12 = circshift(x3_12,1);
x3_12r = real(x3_12);

for n = 1:32 % x3_all
    for k = 1:32 % the positive and negative parts
        x3_all(n) = x3_all(n)+a3(k)*exp(1j*(k-1)*2*pi/32*n);
    end
end
x3_all = circshift(x3_all,1);
x3_allr = real(x3_all);

% plots
subplot(5,1,1); % x3_2[n]
stem(nx,x3_2r);
xlabel('n');
ylabel('x3_2[n]');
axis([0,31,0,1]);

subplot(5,1,2); % x3_8[n]
stem(nx,x3_8r);
xlabel('n');
```

```
157 ylabel('x3_8[n]');
158 axis([0,31,0,1]);
159
160 subplot(5,1,3); % x3_12[n]
161 stem(nx,x3_12r);
162 xlabel('n');
163 ylabel('x3_12[n]');
164 axis([0,31,0,1]);
165
166 subplot(5,1,4); % x3_all[n]
167 stem(nx,x3_allr);
168 xlabel('n');
169 ylabel('x3_all[n]');
170 axis([0,31,0,1]);
```